EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 13

Basic Maths and Calculus

Partial Derivatives and Rates of Change

By - Excellence Programme

18 September 2026

Topics to be covered

01 5-minute chain-rule recap

02 Partial derivatives

03 Exponential chain-rule practice

04 Parametric slope

05 Tangent slope from a graph

06 Velocity and acceleration

07 Second derivative and curvature

Chain rule and partial derivative

chain rule: $d[f(g(x))]/dx = f'(g(x))g'(x)$

parametric: $dy/dx = (dy/dt)/(dx/dt)$

partial derivative: vary one variable; hold the others constant

$$V(x, y, z) \rightarrow \partial V/\partial x, \partial V/\partial y, \partial V/\partial z$$

Partial derivatives in several variables

A scalar quantity V may depend on x, y and z.

$$V = V(x, y, z)$$

Each partial derivative measures change with respect to one selected variable.

$$\partial V/\partial x,\ \partial V/\partial y,\ \partial V/\partial z$$

The remaining variables are treated as constants.

Hold two variables constant

For ∂V/∂x: y and z are constants

For ∂V/∂y: x and z are constants

For ∂V/∂z: x and y are constants

Differentiate only the selected variable; copy every constant factor.

Rough work / extra work

Partial derivatives: monomial

For $V = x^2y^3z^4$, find $\partial V/\partial x$, $\partial V/\partial y$ and $\partial V/\partial z$.

Solution: Partial derivatives: monomial

Step 1: $\partial V/\partial x = 2xy^3z^4$

Step 2: $\partial V/\partial y = 3x^2y^2z^4$

Step 3: $\partial V/\partial z = 4x^2y^3z^3$

Exponential chain rule

For an exponential with an inner function, keep the exponential and multiply by the inner derivative.

If $y = e^u$, then $dy/dx = e^u(du/dx)$

Check the sign of the exponent carefully.

Exponential: negative x

Differentiate $y = e^{-x}$.

Solution: Exponential: negative x

Step 1: Outer derivative remains e^{-x}

Step 2: Inner derivative: $d(-x)/dx = -1$

$dy/dx = -e^{-x}$.

Rough work / extra work

Exponential: negative x^2

Differentiate $y = e^{-x^2}$.

Solution: Exponential: negative x^2

Step 1: Outer derivative: e^{-x^2}

Step 2: Inner derivative: $d(-x^2)/dx = -2x$

$dy/dx = -2x\, e^{-x^2}$.

Rapid review: cosine chain rule

Differentiate $y = A\cos(kx)$, where A and k are constants.

Solution: Rapid review: cosine chain rule

Step 1: $d[\cos(kx)]/dx = -\sin(kx)\cdot k$

$dy/dx = -Ak\sin(kx)$.

Rapid review: e^{3x}

Differentiate $y = 5e^{3x}$.

Solution: Rapid review: e^{3x}

Step 1: $d(e^{3x})/dx = 3e^{3x}$

$dy/dx = 15e^{3x}$.

Rough work / extra work

Exponential: e^{-4x}

Differentiate $y = e^{-4x}$.

Solution: Exponential: e^{-4x}

Step 1: The inner exponent is -4x

Step 2: d(-4x)/dx = -4

dy/dx = $-4e^{-4x}$.

Rapid review: sine chain rule

Differentiate $y = 3\sin(x^2+2)$.

Solution: Rapid review: sine chain rule

Step 1: Outer derivative: $3\cos(x^2+2)$

Step 2: Inner derivative: $2x$

$dy/dx = 6x\cos(x^2+2)$.

Rough work / extra work

Parametric slope

When x and y depend on a parameter, divide their rates with respect to that parameter.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

This gives the tangent slope without first eliminating the parameter.

Parametric practice

If $y = 2t^3+1$ and $x = t^2$, find dy/dx.

Solution: Parametric practice

Step 1: $dy/dt = 6t^2$ and $dx/dt = 2t$

$dy/dx = 3t$, for $t \neq 0$.

Parametric trigonometric slope

If $y = \sin\theta$ and $x = \cos\theta$, find dy/dx.

Solution: Parametric trigonometric slope

Step 1: $dy/d\theta = \cos\theta$

Step 2: $dx/d\theta = -\sin\theta$

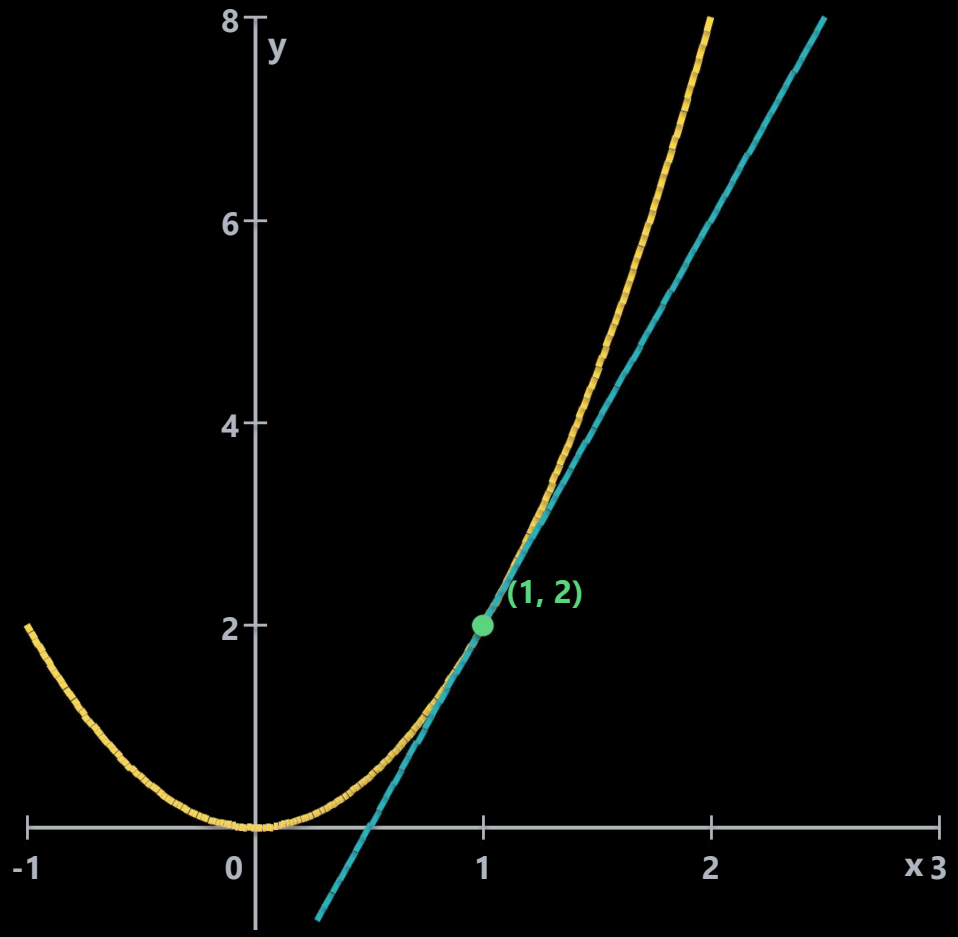
$dy/dx = -\cot\theta$, where $\sin\theta \neq 0$.

Rough work / extra work

Derivative gives the exact tangent slope

For $y = 2x^2$, $dy/dx = 4x$. At $x = 1$, the exact slope is 4.

tangent at (1, 2): y-2 = 4(x-1)



Slope of a parabola

Find the slope of $y = 2x^2$ at $x = 1$.

Solution: Slope of a parabola

Step 1: $dy/dx = 4x$

Step 2: At $x = 1$: slope $= 4(1)$

Slope = 4.

Slope practice

Find the slope of $y = 3x^2+5x-4$ at $x = 1$.

Solution: Slope practice

Step 1: $dy/dx = 6x+5$

At $x = 1$, slope $= 6+5 = 11$.

Rough work / extra work

Power and chain rule together

Differentiate $y = 4/(x^2-5)$, where $x^2 \neq 5$.

Solution: Power and chain rule together

Step 1: Write $y = 4(x^2-5)^{-1}$

Step 2: $dy/dx = 4(-1)(x^2-5)^{-2}(2x)$

$dy/dx = -8x/(x^2-5)^2$.

Negative half-power

Differentiate $y = 3/\sqrt{(x^2-4x)}$, where $x^2-4x > 0$.

Solution: Negative half-power

Step 1: Write $y = 3(x^2-4x)^{-1/2}$

Step 2: $dy/dx = 3(-1/2)(x^2-4x)^{-3/2}(2x-4)$

$dy/dx = -3(x-2)/(x^2-4x)^{3/2}$.

Velocity and acceleration

If position x changes with time t, its first derivative is velocity.

$$v = dx/dt$$

The derivative of velocity is acceleration.

$$a = dv/dt = d^2x/dt^2$$

First derivative: velocity. Second derivative: acceleration.

Rough work / extra work

Second derivative and curvature

The first derivative dy/dx gives slope.

The second derivative d^2y/dx^2 tells how the slope changes with x.

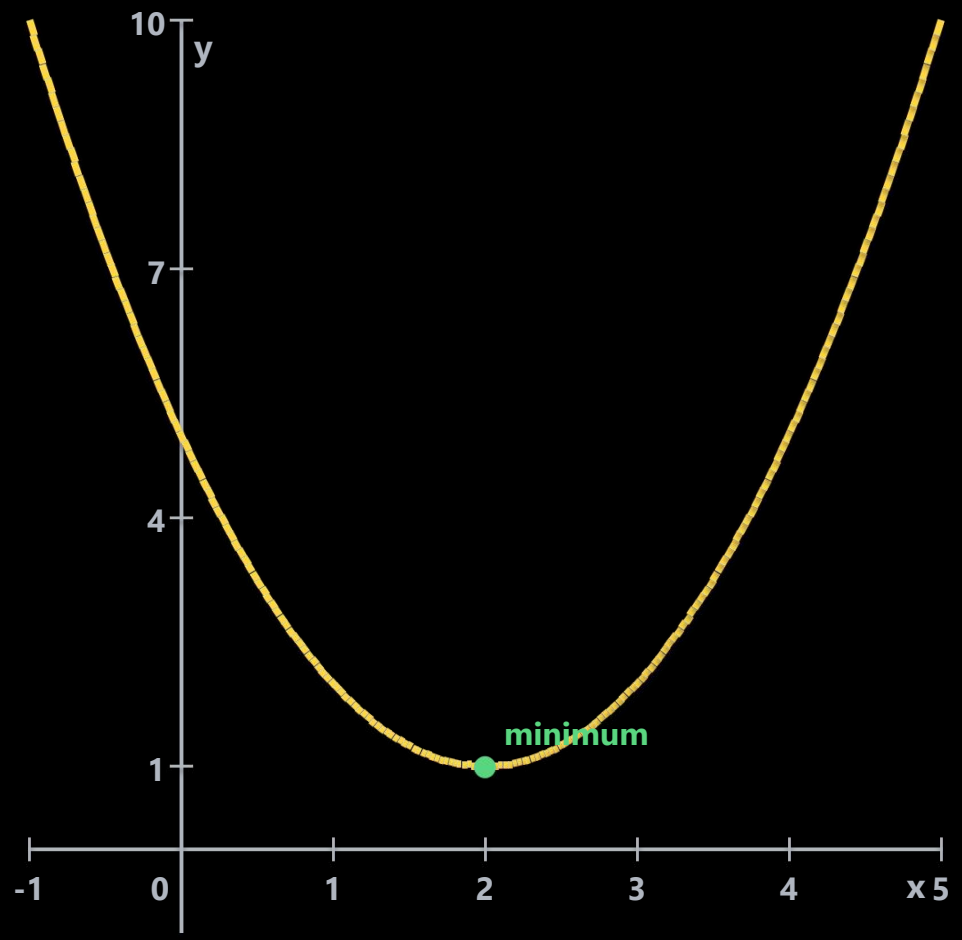
$d^2y/dx^2 > 0$ → slope increases → curve is concave upward

$d^2y/dx^2 < 0$ → slope decreases → curve is concave downward

Increasing slope on a concave-up curve

For $y = (x-2)^2+1$, $dy/dx = 2(x-2)$ and $d^2y/dx^2 = 2 > 0$.

As x increases, slope changes from negative to zero to positive.



Differentiation lock lines

partial derivative: change one variable at a time

chain rule: outer derivative × inner derivative

parametric slope: $dy/dx = (dy/dt)/(dx/dt)$

velocity = dx/dt; acceleration = d^2x/dt^2

second derivative describes change of slope

Practice

1. Redo every partial-derivative question without looking at the answer.

2. Redo the exponential chain-rule examples.

3. Draw $y = 2x^2$ and its tangent at $x = 1$.

4. Explain why velocity and acceleration are derivatives.

Always identify the independent variable before differentiating.